No limit cycle in the Banaji–Boros–Hofbauer three-reaction example

chemistry-open-problems · 18 October 2026 · preliminary

Dulac argument and subfamily: Claude Opus 5.5 exploration agents (scout and verifier, effort high), 17 Oct 2026 · Certificate and manuscript: Claude Opus 5.5 (effort xhigh) · Independent verification: pending

Abstract. Banaji, Boros and Hofbauer classified the three-reaction planar quadratic mass-action networks that admit an Andronov–Hopf bifurcation, and asked whether the networks of their cases 9 and 10 without such a bifurcation can nevertheless have a limit cycle; as an example they asked about the network $2X \to 3X + 2Y,\ X + Y \to 0,\ Y \to X + Y$. We show that this system has no periodic orbit for any rate constants, using a Dulac function that turns the divergence into a negative definite quadratic form. The same argument shows that in the subfamily $2X \to (2 + c_1)X + d_1Y,\ X + Y \to 0,\ Y \to (1 + c_3)X + Y$, periodic orbits can only occur in the Andronov–Hopf region $c_1 < d_1 < 2c_1$.

The question

Banaji, Boros and Hofbauer [1] classify the three-reaction, planar, quadratic mass-action systems that admit an Andronov–Hopf bifurcation (their Theorem 9, cases 9 and 10). In their Conclusions (§6) they write:

> *"However, we leave it open whether the networks in cases 9 and 10 admitting no Andronov–Hopf bifurcations can have a limit cycle. For example, we do not know whether the system $2X \to 3X + 2Y$ (κ_1), $X + Y \to 0$ (κ_2), $Y \to X + Y$ (κ_3), $\dot{x} = \kappa_1 x^2 - \kappa_2 xy + \kappa_3 y$, $\dot{y} = 2\kappa_1 x^2 - \kappa_2 xy$ admits a limit cycle."*

We settle the example (it has no periodic orbit at all, for any rate constants) and, with the same argument, the subfamily of case-9 networks to which it belongs.

Theorem 1. For all $\kappa_1, \kappa_2, \kappa_3 > 0$, the system

$$\dot{x} = \kappa_1 x^2 - \kappa_2 xy + \kappa_3 y, \qquad \dot{y} = 2\kappa_1 x^2 - \kappa_2 xy$$

has no periodic orbit in the closed positive quadrant. In particular it has no limit cycle.

Theorem 2. Let c_1, c_3, d_1 be positive integers and consider the networks

$$2X \xrightarrow{\kappa_1} (2 + c_1)X + d_1Y, \qquad X + Y \xrightarrow{\kappa_2} 0, \qquad Y \xrightarrow{\kappa_3} (1 + c_3)X + Y,$$

i.e. case 9 of Theorem 9 [1] with reaction vectors $c = (c_1, -1, c_3)$, $d = (d_1, -1, 0)$. The mass-action system

$$\dot{x} = c_1\kappa_1 x^2 - \kappa_2 xy + c_3\kappa_3 y, \qquad \dot{y} = d_1\kappa_1 x^2 - \kappa_2 xy$$

has a periodic orbit for some κ only if $c_1 < d_1 < 2c_1$, which is exactly the region where it undergoes an Andronov–Hopf bifurcation Theorem 9 [1]. For $d_1 \le c_1$ or $d_1 \ge 2c_1$ it has no periodic orbit for any $\kappa > 0$.

The example of Theorem 1 is the case $c_1 = c_3 = 1$, $d_1 = 2$, on the boundary $d_1 = 2c_1$ of the Hopf region.

1 Proofs

Proof (of Theorem 2, case $d_1 \ge 2c_1$)**.** Write $F = (f, g)$ for the vector field and take the Dulac function $B(x, y) = y^{-b}$ on $\mathbb{R}^2_{>0}$, with

$$b = 1 + \frac{2\kappa_1(d_1 - c_1)}{\kappa_2}.$$

A direct computation gives

$$y^{b+1}\,\mathrm{div}(BF) = Q(x, y) := \frac{d_1\kappa_1\big(2(c_1 - d_1)\kappa_1 - \kappa_2\big)}{\kappa_2}\,x^2 + 2d_1\kappa_1\,xy - \kappa_2\,y^2,$$

a quadratic form with y^2 coefficient $-\kappa_2 < 0$ and discriminant

$$\Delta = 4d_1\kappa_1\big(\kappa_1(2c_1 - d_1) - \kappa_2\big).$$

(The exponent b minimises Δ among Dulac functions y^{-b}.) If $d_1 \ge 2c_1$, then $\kappa_1(2c_1 - d_1) - \kappa_2 \le -\kappa_2 < 0$, so $\Delta < 0$ and Q is negative definite: $\mathrm{div}(BF) < 0$ on the open quadrant. Since $\mathbb{R}^2_{>0}$ is simply connected, the Bendixson–Dulac criterion excludes periodic orbits contained in it.

On $\{y = 0,\ x > 0\}$ we have $\dot y = d_1\kappa_1 x^2 > 0$, on $\{x = 0,\ y > 0\}$ we have $\dot x = c_3\kappa_3 y > 0$, and the origin is an equilibrium. A periodic orbit in the closed quadrant cannot contain an equilibrium, and cannot touch a boundary point where the field points strictly into the open quadrant (it would have to reach that point from the inside). Hence every periodic orbit would lie in the open quadrant, which is impossible. □

Proof (of Theorem 2, case $d_1 \le c_1$)**.** The stoichiometric matrix $\Gamma = \begin{pmatrix} c_1 & -1 & c_3 \\ d_1 & -1 & 0 \end{pmatrix}$ has rank 2 (its rows are not proportional because $c_3 \neq 0$), so $\ker\Gamma$ is spanned by $u = c \times d = (c_3,\ c_3d_1,\ d_1 - c_1)$. A positive equilibrium x requires a positive flux vector $v(x) \in \ker\Gamma \cap \mathbb{R}^3_{>0}$, i.e. u or $-u$ positive, which fails when $d_1 \le c_1$. So there is no positive equilibrium. A periodic orbit in the open quadrant would enclose an equilibrium (index theory), which would be positive; hence there is none, and the boundary argument above excludes orbits touching the axes. □

Proof (of Theorem 1)**.** This is $c_1 = c_3 = 1$, $d_1 = 2$: $b = 1 + 2\kappa_1/\kappa_2$, $Q = -\frac{2\kappa_1(2\kappa_1 + \kappa_2)}{\kappa_2}\,x^2 + 4\kappa_1 xy - \kappa_2 y^2$ and $\Delta = -8\kappa_1\kappa_2 < 0$. □

Remark 3. In the example, the unique positive equilibrium is $(2\kappa_3/\kappa_2,\ 4\kappa_1\kappa_3/\kappa_2^2)$, with $\det J = 4\kappa_1\kappa_3^2/\kappa_2 > 0$ and $\mathrm{tr}\,J = -2\kappa_3 < 0$.

For case 9 with $c_3 > 0$, Theorem 9 [1] gives an Andronov–Hopf bifurcation iff $\frac{1}{2}\,(d_3/c_3 + d_1/c_1) < d_2/c_2 < d_1/c_1$. In the subfamily, $d_2/c_2 = 1$ and $d_3 = 0$, so the condition reads

$c_1 < d_1 < 2c_1$; for the example, $\frac{1}{2}(0+2) = 1 < 1 < 2$ fails by equality. Theorem 2 shows that in this subfamily "no Andronov–Hopf bifurcation" implies "no periodic orbit". ⌟

2 What remains open

The general question of [1], for all case-9 and case-10 networks without an Andronov–Hopf bifurcation (case 9 with $d_2 \neq -1$ or $d_3 \neq 0$, including the branch $c_3 = 0$, and all of case 10), is not settled here. The same strategy with Dulac functions $x^\alpha y^\beta$ reduces it to sign conditions on a cubic-plus-quadratic form.

3 Computer verification

`scripts/certify_dulac.py` checks every identity above symbolically (sympy): the Dulac computation, the coefficients and discriminant in the subfamily and in the example, the vector $u = c \times d$, the equilibrium and its Jacobian, and the boundary behaviour.

References

[1] Murad Banaji, Balázs Boros and Josef Hofbauer. Oscillations in three-reaction quadratic mass-action systems. *Studies in Applied Mathematics*, **2024**.
DOI: 10.1111/sapm.12639 ARXIV: 2304.02303 1, 2, 3